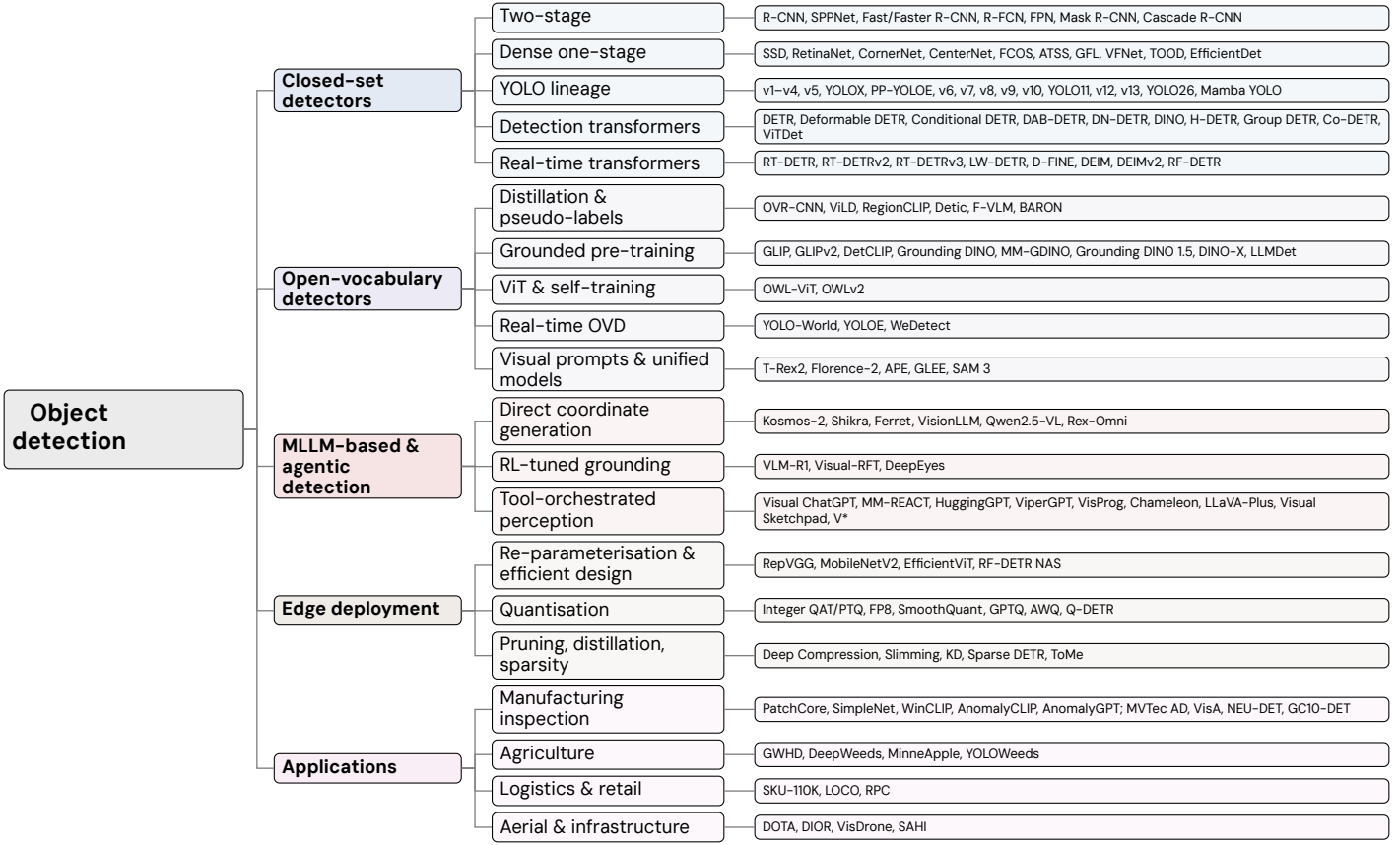
Object detection
Closed-set detectors
Two-stage
R-CNN, SPPNet, Fast/Faster R-CNN, R-FCN, FPN, Mask R-CNN, Cascade R-CNN
Dense one-stage
SSD, RetinaNet, CornerNet, CenterNet, FCOS, ATSS, GFL, VFNet, TOOD, EfficientDet
YOLO lineage
v1–v4, v5, YOLOX, PP-YOLOE, v6, v7, v8, v9, v10, YOLO11, v12, v13, YOLO26, Mamba YOLO
Detection transformers
DETR, Deformable DETR, Conditional DETR, DAB-DETR, DN-DETR, DINO, H-DETR, Group DETR, Co-DETR, ViTDet
Real-time transformers
RT-DETR, RT-DETRv2, RT-DETRv3, LW-DETR, D-FINE, DEIM, DEIMv2, RF-DETR
Open-vocabulary detectors
Distillation & pseudo-labels
OVR-CNN, ViLD, RegionCLIP, Detic, F-VLM, BARON
Grounded pre-training
GLIP, GLIPv2, DetCLIP, Grounding DINO, MM-GDINO, Grounding DINO 1.5, DINO-X, LLMDet
ViT & self-training
OWL-ViT, OWLv2
Real-time OVD
YOLO-World, YOLOE, WeDetect
Visual prompts & unified models
T-Rex2, Florence-2, APE, GLEE, SAM 3
MLLM-based & agentic detection
Direct coordinate generation
Kosmos-2, Shikra, Ferret, VisionLLM, Qwen2.5-VL, Rex-Omni
RL-tuned grounding
VLM-R1, Visual-RFT, DeepEyes
Tool-orchestrated perception
Visual ChatGPT, MM-REACT, HuggingGPT, ViperGPT, VisProg, Chameleon, LLaVA-Plus, Visual Sketchpad, V*
Edge deployment
Re-parameterisation & efficient design
RepVGG, MobileNetV2, EfficientViT, RF-DETR NAS
Quantisation
Integer QAT/PTQ, FP8, SmoothQuant, GPTQ, AWQ, Q-DETR
Pruning, distillation, sparsity
Deep Compression, Slimming, KD, Sparse DETR, ToMe
Applications
Manufacturing inspection
PatchCore, SimpleNet, WinCLIP, AnomalyCLIP, AnomalyGPT; MVTec AD, VisA, NEU-DET, GC10-DET
Agriculture
GWHD, DeepWeeds, MinneApple, YOLOWeeds
Logistics & retail
SKU-110K, LOCO, RPC
Aerial & infrastructure
DOTA, DIOR, VisDrone, SAHI